

PIXOUS TECHNOLOGIES

Unit Testing Specification

HR Management Portal — Client Submission

Version 1.0 · 2026-08-14

Pixous HR Portal — Unit Testing Document

Field	Value
Version	1.0
Date	2026-08-14
Backend framework	JUnit 5 + Mockito (Maven, <code>mvn test</code>)
Frontend framework	Vitest + React Testing Library + jsdom (<code>npm test</code>)
Current inventory	62 automated tests: 40 backend + 22 frontend — all passing
Coverage targets	Backend service layer ≥ 70%; frontend critical paths 100%

1. Testing Strategy

1.1 Test Pyramid

E2E (Playwright)	<- planned, top-10 journeys
Integration (Spring slices)	<- planned, service + repository
UNIT + Component [THIS DOC]	<- 62 tests, the foundation

Unit tests are fast, deterministic, dependency-free (repos/interfaces mocked), and run in seconds. They are the gate for every commit: `mvn test` and `npm test` must be green before any release.

1.2 How to Run

```
# Backend (all 40 tests)
cd backend
mvn test

# Single class
mvn test -Dtest=JwtServiceTest

# Frontend (all 22 tests)
cd web
npm install      # first time only
npm test        # = vitest run

# Watch mode (development)
npx vitest
```

1.3 Conventions

- File layout mirrors `src/main`: `backend/src/test/java/...`, `web/src/**/*.test.ts(x)`.
- Naming: `methodName_scenario_expectedResult` (e.g. `login_wrongPassword_throws401`).
- One behaviour per test; assert on behaviour (exceptions, return values), not implementation.
- Mock every external dependency (repositories, file system, HTTP).
- Frontend component tests never hit the network — the API module is mocked.

2. Backend Unit Tests (40)

2.1 `security/JwtServiceTest` — JWT core

#	Test case	Input	Expected
B-01	<code>generatesTokenThatParsesBackToSameIdentity</code>	username, roles	Token round-trips to identical subject/roles
B-02	<code>tokenExpiresAtConfiguredTtl</code>	TTL 30 s	After expiry, validation fails
B-03	<code>rejectsTamperedPayload</code>	Token with payload edited	Signature mismatch → rejected
B-04	<code>rejectsTokenSignedWithDifferentSecret</code>	Token from other key	Rejected
B-05	<code>rejectsGarbageAndEmptyTokens</code>	"abc" , "" , null	Rejected
B-06	<code>missingRolesClaimYieldsEmptyList</code>	Token without roles claim	Empty role list, no exception

2.2 `security/LoginAttemptLimiterTest` — brute-force guard

#	Test case	Input	Expected
B-07	<code>allowsAttemptsUpToThreshold</code>	1–7 failures	Not blocked
B-08	<code>blocksAfterEightFailures</code>	8th failure	Blocked (lockout)
B-09	<code>blocksEitherIdentityIndependently</code>	lock user A	B unaffected
B-10	<code>successfulLoginClearsTheStreak</code>	5 fails + success	Streak reset, not blocked
B-11	<code>countingIsCaseInsensitiveOnUsername</code>	Admin vs admin	Same counter
B-12	<code>differentUsersAreNotBlockedByEachOther</code>	users A, B	Isolation confirmed

2.3 `common/StorageServiceTest` — file upload hardening

#	Test case	Input	Expected
B-13	<code>keepsAllowlistedExtension</code>	report.pdf , photo.png	Stored with original extension
B-14	<code>htmlUploadIsStoredAsBin</code>	page.html	Stored as .bin
B-15	<code>svgUploadIsStoredAsBin</code>	logo.svg	Stored as .bin (XSS guard)
B-16	<code>doubleExtensionTakesLastPart</code>	evil.jpg.exe	Evaluated on final extension
B-17	<code>uppercaseExtensionIsNormalized</code>	PHOTO.PNG	Normalised, allowed
B-18	<code>noExtensionBecomesBin</code>	README	Stored as .bin
B-19	<code>emptyFileIsRejected</code>	0-byte upload	Rejected
B-20	<code>uploadedFilenameIsNeverUsedInPath</code>	../../etc/passwd name	Path is generated, traversal impossible

2.4 `modules/auth/AuthServiceTest` — authentication flows

#	Test case	Input	Expected
B-21	loginRequiresBothFields	Missing username/password	Validation failure
B-22	loginAcceptsValidCredentials	Correct credentials	Success
B-23	loginSuccessIssuesTokensAndClearsFailures	Correct after failures	Tokens issued; failure streak cleared
B-24	loginWithUnknownUserThrowsBadCredentialsAndRecordsFailure	Unknown user	401-style error; failure recorded
B-25	loginByFullNameWorksWhenSingleMatch	Full name of unique user	Login succeeds
B-26	fiveWrongPasswordsLockTheAccount	5 wrong attempts	Account locked
B-27	disabledAccountRejectedEvenWithCorrectPassword	Disabled + correct password	Rejected
B-28	legacyUserWithoutCompanyIsAutoAssignedToPixious	User with null company	Assigned to default tenant
B-29	refreshRotatesTheTokenAndRevokesTheOldOne	Valid refresh token	New pair issued; old revoked
B-30	refreshWithRevokedTokenIsRejected	Revoked token	Rejected
B-31	refreshWithExpiredTokenIsRejected	Expired token	Rejected
B-32	changePasswordRejectsWrongCurrentPassword	Wrong current password	Rejected
B-33	changePasswordSuccessRevokesAllSessions	Correct current + new	All old sessions revoked
B-34	signupRejectsDuplicateUsername	Existing username	Rejected
B-35	signupAssignsDefaultRoleAndEncodesPassword	Valid signup	Default role + hashed password
B-36	changePasswordRequiresNewPasswordAtLeastEightChars	7-char password	Rejected
B-37	signupRejectsShortUsernameAndPassword	Short values	Rejected
B-38	signupRejectsMalformedAadhaar	Bad Aadhaar format	Rejected
B-39	signupRejectsMalformedPhone	Bad phone format	Rejected
B-40	signupAcceptsValidPayload	Fully valid payload	Accepted

3. Frontend Unit & Component Tests (22)

3.1 `src/lib/utils.test.ts` — shared utilities

#	Test case	Input	Expected
F-01	merges and dedupes tailwind classes	<code>cn("px-2","px-4")</code>	<code>"px-4"</code>
F-02	filters falsy values	<code>cn("a",null,false,"b")</code>	<code>"a b"</code>
F-03	handles empty input	<code>initials("")</code>	<code>"?"</code>
F-04	takes first letters of up to two words	<code>"Sethuraman Balasubramanian"</code>	<code>"SB"</code>
F-05	uppercases and skips empty tokens	<code>" john doe "</code>	<code>"JD"</code>
F-06	maps 1..12 to English month names	<code>monthName(1), (12)</code>	<code>"January", "December"</code>
F-07	falls back to the number for out-of-range values	<code>monthName(0), (13)</code>	<code>"0", "13"</code>
F-08	formats INR with Indian digit grouping	<code>formatMoney(100000)</code>	<code>"₹1,00,000.00"</code>
F-09	accepts numeric strings and non-finite values	<code>"5000", undefined</code>	<code>"₹5,000.00", "₹0.00"</code>
F-10	formats whole hours and minutes	<code>minutesToHours(90)</code>	<code>"1h 30m"</code>
F-11	returns em dash for missing or non-positive values	<code>null, 0, -5</code>	<code>"—"</code>

3.2 `src/lib/dates.test.ts` — date/time helpers

#	Test case	Input	Expected
F-12	converts 24-hour times to 12-hour with AM/PM	<code>"14:30", "09:05", "00:00", "12:00"</code>	<code>"2:30 PM", "9:05 AM", "12:00 AM", "12:00 PM"</code>
F-13	pads single-digit minutes	<code>"08:5"</code>	<code>"8:05 AM"</code>
F-14	returns em dash for empty input	<code>undefined, null, ""</code>	<code>"—"</code>
F-15	returns input untouched when it is not parseable	<code>"banana"</code>	<code>"banana"</code>
F-16	returns today in <code><input type=date></code> format	—	matches <code>\d{4}-\d{2}-\d{2}</code>
F-17	returns this month in <code><input type=month></code> format	—	matches <code>\d{4}-\d{2}</code>

3.3 `src/pages/Login.test.tsx` — login page component

#	Test case	Setup	Expected
F-18	renders the sign-in form	Render page	Heading, username/password fields, submit button present

F-19	shows validation errors when submitting empty fields	Click submit empty	"Username is required" + "Password is required"; login NOT called
F-20	calls login and navigates home on valid credentials	Type + submit	<code>login("admin", "Test1234@")</code> called; navigate to /
F-21	does not navigate when login fails, keeps form usable	Mock login rejects	No navigation; button re-enabled for retry
F-22	toggles password visibility	Click eye button	Input type <code>password</code> ↔ <code>text</code>

4. Test Data & Fixtures

Concern	Approach
Backend	In-memory mocks for repositories; fixed usernames/passwords; no DB required
Frontend	<code>localStorage.clear()</code> before each test; <code>useAuth</code> and router stubbed; API mocked
Determinism	Vitest <code>globals: false</code> ; explicit RTL cleanup in <code>src/test/setup.ts</code>
Secrets	Test files never contain real credentials (placeholders only)

5. CI Integration

Local gate: `mvn test` + `npm test` must pass before push (watcher auto-pushes on save).

Docker build: backend image builds with `-DskipTests` (fast deploys); tests are the developer/PR gate.

Recommended next step: add a `test` job to `.github/workflows/deploy.yml` or a separate CI workflow that fails the build on test failure.

6. Coverage Report

```
# Backend line coverage (add jacoco to pom, then:)
mvn test jacoco:report
# -> target/site/jacoco/index.html

# Frontend coverage
cd web && npx vitest run --coverage
```

Targets: backend service layer ≥ 70% line coverage; frontend critical flows (auth, uploads, payroll) 100%.

7. Backlog — Next 90 Test Cases (prioritised)

Priority	Area	Suggested cases
P1	Company/Tenant service	Tenant creation, code generation, uniqueness, cross-tenant isolation (2 tenants in 1 test)
P1	Leave service	Balance validation, overlap rejection, approval state machine, policy accrual
P1	Payroll service	Run generation, duplicate-period guard, payslip totals/rounding
P1	File controller	Unauthenticated access, upload validation, traversal via URL
P2	Attendance service	Clock-in/out, duplicate shift, grace rules

P2	User service	Pagination, filters, role assignment, self-demotion guard
P2	Frontend: API client	Token attach, 401 → refresh retry, refresh failure → logout
P2	Frontend: forms	Employee, leave, claim, payslip filter (validation + submit + toast)
P2	Notification/Audit service	Fan-out, unread counts, audit entries by actor/action/date
P3	Frontend: data tables	Pagination, sorting, search debounce, empty state
P3	Chat/WS	Message send, reconnect (mock STOMP)

8. Definition of Done

- [] All 62 existing tests still pass
- [] New code ships with at least one unit test per behaviour (positive + negative)
- [] No test uses real credentials or network access
- [] `mvn test` and `npm test` green on the machine before push
- [] Coverage never regresses below current level